Machine Learning Surrogates for Mudflow Depth Profiles under Varying Yield Stress

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Machine Learning Mini-Project | https://github.com/jefffei0314-arch/mudflow-ml-surrogate

1 Problem and Motivation

Mudflow behaviour depends on rheological properties, particularly yield stress, which affects flow resistance, depth and propagation extent. Tan et al. [1] previously investigated the effects of rheological models and parameters on mudflow propagation using a depth-averaged numerical model. Their findings demonstrated that rheological conditions influence the resulting flow-depth profiles.

Repeated numerical simulations under different conditions can be computationally demanding. This project investigates the ability of machine-learning (ML) surrogate models to accurately predict flow-depth profiles at unseen yield-stress conditions using data generated from numerical simulations. The objective is to evaluate flow-depth prediction accuracy and identify limitations in reproducing sharp flow fronts.

2 Dataset and Method

Dataset. The data originate from the one-dimensional, depth-averaged dam-break simulations in Tan et al. [1], specifically the Quadratic rheological model results in their Fig. 10(b).

The dam-break problem represents the propagation of an initially confined mudflow mass after release. As specified in Table 2 of that study, the initial mudflow width is $x_0 = 305\,\mathrm{m}$ and its initial depth is $h_0 = 30.5\,\mathrm{m}$.

Nine yield stresses (20, 50, 80, 200, 500, 800, 1000, 2000 and 3000 Pa) were considered. The flow-depth profiles were extracted at a simulation time of $t = 180\,\mathrm{s}$ for nine different yield-stress conditions. Each profile contains 35,000 spatial data points, resulting in a total of 315,000 data samples for ML model development.

The ML inputs are normalized position x/x_0 and yield stress τ_y; the target is normalized flow depth h/h_0:

$$\frac{\hat{h}}{h_0} = f_{\mathrm{ML}}\left(\frac{x}{x_0}, \tau_y\right), \qquad x_0 = 305\,\mathrm{m}, \quad h_0 = 30.5\,\mathrm{m}. \tag{1}$$

Models. Two scikit-learn [2] regressors were considered: Polynomial Ridge Regression, using standardized inputs, polynomial features and L2 regularization, and Random Forest Regression, which combines decision trees to capture nonlinear relationships. Negative predicted depths were clipped to zero.

Validation and Hyperparameter Selection. Yield stresses of 50, 500 and 2000 Pa were selected as held-out test cases in three separate experiments. For each experiment, the remaining eight profiles (280,000 points) formed the training dataset. Hyperparameters were selected using grid search with eight-fold group-aware cross-validation (`GroupKFold`), validating on one complete yield-stress profile per fold. Polynomial degrees 2–4 and Ridge regularization parameters $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ were tested. Random Forest used 100 or 200 trees, maximum depth 8 or 12, and minimum leaf size 1 or 3. The configuration with the lowest mean validation MSE was selected.

Model Refitting and Testing. Each selected model was refitted using all eight training profiles and evaluated on the excluded test profile (35,000 points). Prediction accuracy was assessed using R^2, RMSE and MAE, with depth errors reported in metres.

3 Results and Discussion

Table 1: Group-aware validation results and selected hyperparameters. CV RMSE is the square root of mean fold MSE.

Test (Pa)	Model	Selected hyperparameters	CV RMSE (m)
50	Ridge	Degree 4, $\alpha = 1$	0.6485
50	Forest	200 trees, depth 12, leaf 1	0.7676
500	Ridge	Degree 4, $\alpha = 10$	0.6436
500	Forest	200 trees, depth 12, leaf 1	0.7088
2000	Ridge	Degree 3, $\alpha = 100$	0.8168
2000	Forest	100 trees, depth 12, leaf 1	0.8321

Table 2: Final performance on independent held-out yield-stress profiles.

τ_y (Pa)	Model	R^2	RMSE (m)	MAE (m)
50	Polynomial Ridge	0.7657	0.4315	0.3408
50	Random Forest	**0.9711**	**0.1514**	**0.0440**
500	Polynomial Ridge	**0.9024**	**0.4569**	**0.3044**
500	Random Forest	0.7053	0.7942	0.4382
2000	Polynomial Ridge	**0.8917**	**0.8915**	**0.7171**
2000	Random Forest	0.7454	1.3665	0.9409

Table 1 presents the optimal hyperparameter configurations selected by minimizing the mean cross-validation MSE, with validation errors reported as RMSE. These configurations were subsequently used to refit both models before evaluating their performance on the held-out test profiles. Table 2 presents the final testing performance of Polynomial Ridge and Random Forest on the three held-out yield-stress profiles, evaluated using R^2, RMSE and MAE.

At 50 Pa, Random Forest outperformed Polynomial Ridge in all three evaluation metrics, whereas Polynomial Ridge achieved higher R^2 and lower RMSE and MAE at 500 and 2000 Pa (Table 2). However, Fig. 1 shows that Random Forest reproduced the mudflow profile shape more closely, while Polynomial Ridge produced overly smooth profiles. This difference may be explained by their modelling approaches: Polynomial Ridge uses a global polynomial function that struggles to represent abrupt changes in flow depth, whereas Random Forest can capture localized variations through its decision-tree structure.

The numerical simulations produce shorter runout distances than the profiles predicted by Random Forest. This discrepancy may be related to the yield-stress resistance term in the Quadratic rheological model, $S_{f,y} = \tau_y/(\rho g h)$ [1], which increases as flow depth decreases near the advancing front, limiting further propagation. Since Random Forest does not explicitly account for this physical relationship, it may predict positive depths beyond the numerical flow front, contributing to larger errors where the reference depth is zero. In contrast, the overly smooth Polynomial Ridge predictions make it difficult to identify a distinct flow front, despite achieving lower overall errors at 500 and 2000 Pa. Fig. 2 further shows that Random Forest exhibits sharp positive error peaks near the numerical flow front, particularly at 500 and 2000 Pa, whereas Polynomial Ridge produces smoother, more spatially distributed errors. These findings suggest that physics-guided learning may improve the physical consistency of future mudflow predictions.

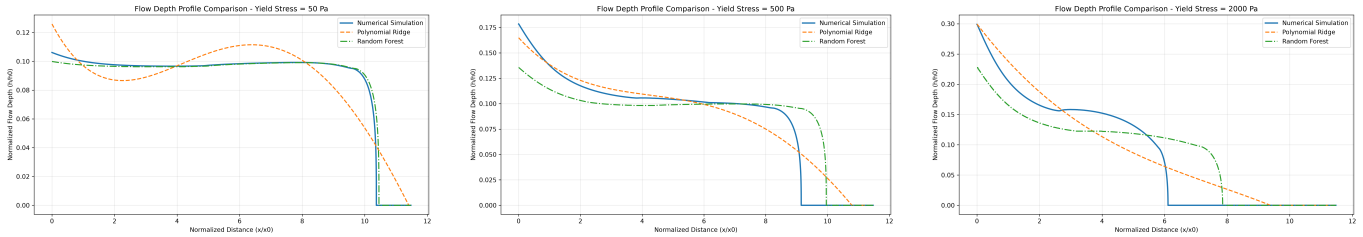


Figure 1: Flow-depth profiles from numerical simulations and ML predictions at 180 s for 50, 500 and 2000 Pa (left to right).

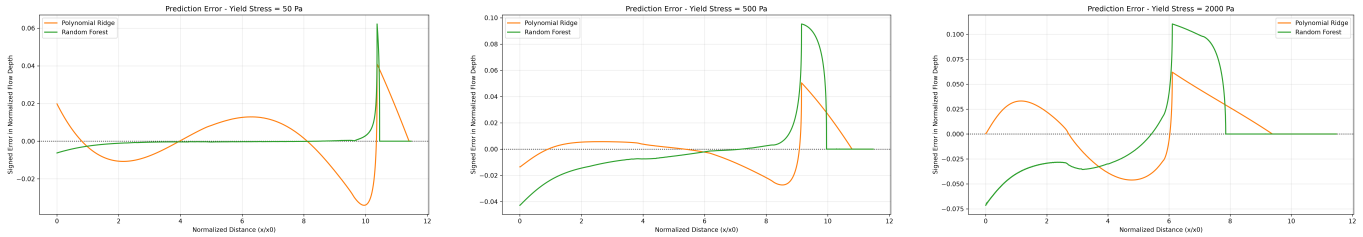


Figure 2: Signed normalized-depth errors between ML predictions and numerical simulations at 50, 500 and 2000 Pa (left to right).

4 Conclusion and Reflection

Although the dataset contains 315,000 spatial observations, these originate from only nine yield-stress conditions. Group-aware validation reduces data leakage, but the held-out tests assess only interpolation within the sampled stress range (20–3000 Pa). Furthermore, neither ML model explicitly enforces the governing flow equations, limiting confidence in their physical consistency. The results demonstrate the potential of ML surrogates but do not establish their ability to replace the numerical model. Future work should include more diverse simulation conditions, quantitative runout-error assessment and physics-guided learning.

Reproducibility and AI Disclosure

The linked GitHub repository provides Python code, datasets, validation and test results, plots, and reproduction instructions. ChatGPT (OpenAI) assisted with Python implementation, debugging, and LaTeX formatting. The author provided the numerical simulation data, ran the experiments, reviewed the results, drafted the report, and interpreted their engineering significance.

References

[1] J. F. Tan, P. H. Tion, and S. C. Lau, "Parametric Sensitivity Analyses of Mudflow Rheology with Depth-Averaged Model," *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences*, vol. 131, no. 2, pp. 78–94, 2025.

[2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.